Curriculum Vita

Name: Wei-Tao Lyu

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Academic Qualifications

- **01/09/2011 – 01/06/2015**: **Bachelor of Science (BS)** in Physics, School of Physics, Huazhong University of Science and Technology.
- **01/09/2015 – 01/12/2021**: **Doctor of Philosophy (PhD)** in Astronomy, School of Astronomy and Space Science, University of Science and Technology of China.
  Advisor: Prof. Sheng-Cai Shi.

Academic Positions

- **01/01/2022 – 30/07/2022**: **Research Associate**, Department of Physics, Chinese University of Hong Kong. **Supervisor:** Prof. Hua-Bai Li.
- **03/08/2022 – Present**: **Postdoctoral Fellow**, Department of Physics, Chinese University of Hong Kong. **Supervisor:** Prof. Hua-Bai Li.

Research Expertise

- Design and characterization of **Microwave Kinetic Inductance Detectors (MKIDs)**.
- Development and optimization of readout systems for **Microwave Kinetic Inductance Detectors (MKIDs)**.
- Cryogenic system integration for submillimetre astronomical cameras.

Selected Publications

1. **Lyu, W.**, Li, H. B., Zhi, Q., Li, J., & Shi, S. (2026). Laser-Based Frequency-to-Pixel Mapping Method for Submillimeter MKID Array. RAS Techniques and Instruments, rzag007.

2. **Lv, W.T.**, Ding, J.Q., Zhi, Q., Wang, Z., Li, J., & Shi, S.C. (2021). “Cavity and ground effects on a superconducting microstrip resonator.” *AIP Advances*, 11(9), 095308.
3. **Lyu, W.T.**, Zhi, Q., Hu, J., Li, J., & Shi, S.C. (2024). “Disorder effects in NbTiN superconducting resonators.” *Chinese Physics B*, 33(2), 027401.
4. **Lv, W.T.**, Hu, J., Yang, J.P., Wang, Z., & Shi, S.C. (2018). “Characterization of Nb and TiN superconducting CPW lines with Thru-line calibration method.” *Infrared, Millimeter-Wave, and Terahertz Technologies V*, Vol. 10826. International Society for Optics and Photonics.
5. **Lv, W.T.**, Zhou, L.S., Li, J., Wang, Z., & Shi, S.C. (2021). “Characterization of distributed and lumped-element THz MKIDs.” *AOPC2021: Novel technologies and instruments for astronomical multi-band observations*. International Society for Optics and Photonics.

Research Projects

- **Hong Kong’s Window on the Universe: Building a Pioneering Submillimetre Astronomical Camera**
  - **Role:** Responsible for cryostat system integration with superconducting MKIDs and cold optics.

Technical Skills

- **Instrumentation:** Design and characterization of superconducting resonators and MKIDs.
- **Cryogenics:** Cryostat system integration and optimization for submillimetre astronomy.
- **Data Analysis:** Advanced signal processing and calibration techniques for superconducting devices.
- **Software:** Proficient in simulation and data analysis tools for astronomical instrumentation.

Languages

- **English:** Fluent (reading, writing, speaking).
- **Chinese (Mandarin):** Native proficiency.